MIL-C-82686
20 September 1979
SUPERSEDING
MIL-C-82678(OS)
24 February 1978
and
Amendment 2
11 October 1978

MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT: CKU-7/A REWORK ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the refurbishment of M38 and CKU 7/A rocket catapults which includes renovation of salvageable hardware, regraining and replacement of non-repairable components.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-14814	Cartridge, Impulse, M270
MIL-S-48567	Sleeve and Grain Assembly
MIL-C-48568	Catapult, Aircraft Ejection Seat, CKU-7/A Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Variables
MIL-STD-413	Visual Inspection Guide for Rubber O-rings
MIL-STD-453	Inspection, Radiographic
MS17983	Compass-Magnetic, Pilots Standby

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD (Code Ident. 14083)

PL 11820362	Catapult, Aircraft Ejection Seat, CKU 7/A Metal Parts Assembly
8594155	Pellet
10533785	Lug
10551431	Cartridge, Impulse, M270 Assembly
11820361	Catapult, Aircraft Ejection Seat, CKU 7/A Assembly
11820362	Catapult, Aircraft Ejection Seat, CKU 7/A Metal Parts Assembly
11820365	Sleeve and Grain Assembly

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NAVAL SEA SYSTEMS COMMAND (Code Ident. 53711)

5184309	Retainer
5184310	Tube, Booster
5184311	Shipping Container for Rocket Catapults
5184312	Shipping Container, Inner Packing for Rocket Catapults

(Documents referred to but not listed individually are listed on PL 11820362.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1-78	Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 Conformance to documents. Materials, parts and assemblies, reclaimed and new, shall comply with the requirements specified on PL 11820362, applicable drawings, and referenced documents.

3.2 Catapult receipt, inspection, and disposition. The following procedure shall be followed:

3.2.1 Government furnished material (GFM) shall be inspected during the disassembly process to ascertain reclaimable components are not damaged.

3.2.2 The components listed in TABLE I are expected to have a 95 per cent recovery rate and be retained for modification and/or reuse.

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TABLE I. Reusable components.

Component	Number Required	Drawing/Part Number
Pin, Firing	1	8596909
Key	2	10533783
Lug	2	10533785
Head	1	10551412
Mount	1	10551413
Nozzle	1	10551414
Retainer	1	10551418
Sleeve	1	10551420
Tube, Booster	1	10551422
Tube, Launcher	1	10551423
Tube, Motor	1	10551424
Plate	1	10551433
Track	1	10551421

3.2.3 The reworked catapult assembly shall contain new components as listed in TABLE II:

TABLE II. Required new components.

Component	Number Required	Reference
Ring, Retaining	1	MS16625-1250
O-Ring	1	MS29513-012
O-Ring	1	MS29513-120
O-Ring	1	MS29513-117
O-Ring	2	MS29513-020
O-Ring	2	MS29513-148
O-Ring	1	MS29513-223
O-Ring	2	MS9021-34
O-Ring	1	MS9021-36
Setscrew	2	MS51963-35
Setscrew	1	MS51963-20
Setscrew	4	10551585
Pin, Shear	1	10551415
Washer	1	11820367
Pin, Spring	1	MS9048-104
Spring, Locking	1	10533782
Spring, Grain	1	11731754
Wire, Lock	2	10533781
Ring, Back-up	1	5184306
Release	1	10551416
Cam	1	10551411
Screw	2	11731753
Plug, Protective	1	MS5501/10-F4

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3.2.3.1 Release. Replace the release, which has a 0.14 inch (in.) wide x 0.826 in. diameter O-ring groove in accordance with Dwg. 10551416, revision B, with a release which has a 0.17 in. wide x 0.831 in. diameter O-ring groove in accordance with Dwg. 10551416. Releases which have 0.17 in. wide X 0.831 in. diameter O-ring groove, and conform to Dwg. 10551416, may also be used in the rework assembly.

3.2.3.2 Cam. Replace the cams having a 1/16 in. radius in accordance with Dwg. 10551411, revision D with cams having a 1/4 in. radius in accordance with Dwg. 10551411. Cams which have ¼ in. radius and conform to Dwg. 10551411, may also be used in the rework assembly.

3.2.3.3 Cartridge assembly. Only new M270 impulse cartridges manufactured within 12 months prior to the contract scheduled delivery date for the catapult assembly lot in accordance with Dwg. 10551431, and comply with the requirements of MIL-C-14814, shall be installed in the CKU 7/A rework assembly, Dwg. 11820361.

3.2.3.4 Propellant. Only new sleeve and grain assemblies manufactured within 12 months prior to the contract scheduled delivery date for the catapult assembly lot in accordance with Dwg. 11820365, and conform to the requirements of MIL-S-48567, shall be installed in the CKU-7/A rework assembly, Dwg. 11820361.

3.2.4 Modified components. Conversion of M38 and CKU-7/A catapults to the latest CKU-7/A catapult configuration shall be accomplished by modifying the following components:

3.2.4.1 Booster tube. Modify the booster tube which has a 0.995 in. outside diameter and single O-ring groove configuration in accordance with Dwg. 10551422, Revision C, to the dual O-ring groove configuration in accordance with Dwg. 5184310. Booster tubes which have dual O-ring grooves and conform to Dwg. 10551422 may also be used in the rework assembly.

3.2.4.2 Launcher tube. Modify the launcher tube in accordance with Dwg. 10551423 by eliminating the two nylon pellets, Dwg. 8594155.

3.2.4.3 Track. Modify the track in accordance with Dwg. 10551421 by machining cross sections A-A, B-B, C-C, and D-D to allow for teflon finish.

3.2.4.4 Retainer. Modify the retainer which has a 3/8 in. radius in zone D-7 in accordance with Dwg. 10551418, Revision C, to the configuration shown of Dwg. 5184309. Retainers which have 3/16 and 1/4 in. radii in zone D-7 and conform to Dwg. 10551418 may also be used in the rework assembly.

3.3 Hydrostatic pressure. The components listed in TABLE III shall withstand the corresponding internal pressure for 15 seconds (s) minimum, without application of any external load, and shall thereafter show no evidence of leakage or mechanical failure.

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TABLE III. Internal pressure requirements.

Component	Hydrostatic pressure psi, minimum
Booster tube (10551422)	11,000
Launcher tube (10551423)	11,000
Motor tube (10551424)	5,000

3.4 Torque. Torque shall be as specified in TABLE IV. No loosening or deformation shall occur upon application of breakaway torque.

TABLE IV. Torque values.

Components	Seating or bottoming (Inch-pounds)	Breakaway (Inch-pounds) (Minimum)
Booster tube assembly 10551422 to Head 10551412	400 + 50 - 0	200
Launcher tube assembly 10551423 to Mount 10551413	400 + 50 - 0	200

3.5 Residual magnetism. The assembled catapult shall not deflect the indicator of a MS17983 compass more than 5° in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Nozzle leak test. Nozzle ball/retainer O-ring shall show no indications of leakage throughout the operational range of the swivel nozzle.

3.7 Release O-ring leak test. Release O-ring shall show no indication of leakage when subjected to test pressure.

3.8 Shear pin strength. Shear pins shall shear when subjected to a radial load of 41 ± 5 lbs. applied at the rate of 0.125 inches per minute.

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3.9 Protective finish. Protective finish shall be in accordance with the applicable specification referenced on the drawings and shall apply to all surfaces regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.4).

3.10 Workmanship. The requirements of workmanship shall be as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shots, inclusions, porosity or other metal defects.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part of the assembly shall contain chips, dirt, oil, rust, corrosion or other foreign matter.

3.11 Ballistics. The catapult, loaded with acceptable explosive components, shall function without mechanical failure and conform to the requirements of MIL-C-48568.

3.12 Radiographic requirements. Each loaded rocket catapult shall be radiographically inspected for the presence, condition, and correct placement of internal components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The requirements specified herein are for quality conformance inspection (see 4.4).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: Not applicable
d. Humidity: Room ambient to 95 percent relative, maximum

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4.4 Quality conformance inspection. The quality conformance inspection shall consist of the examinations of 4.5 and the tests of 4.6 performed on samples selected in accordance with 4.4.2 and 4.4.3, respectively. Unless otherwise specified, sampling shall be in accordance with MIL-STD-105.

4.4.1 Lot size. For the purpose of identification and shipment, a shipping lot shall consist of 1200 metal parts assemblies, maximum, exclusive of test samples, provided the homogeneity of the manufacturing process has been retained throughout the inspection lot unless otherwise specified (see 6.2).

4.4.1.1 Identification marking for head. Unless otherwise specified (see 6.2), each head shall have the previous serial number stamped out and a new serial number engraved on either side of the head.

4.4.2 Sampling for examinations. One hundred percent examination shall be performed for all critical defects. Sampling for major and minor defects shall be in accordance with TABLE V.

TABLE V. Sampling plan and individual defect criteria.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 -22000	300	4	5	5	6

4.4.3 Sampling for tests. Sampling for tests shall be in accordance with the following:

4.4.3.1 Hydrostatic pressure. Each component listed in TABLE III shall be subjected to the internal hydrostatic pressure test of 4.6.1.

4.4.3.2 Shear pin strength. Fifty pins from each lot of shear pins used in the contract shall be tested for shear strength in accordance with 4.6.2.

4.4.3.3 Protective finish. The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable documents listed on the drawings (see 4.6.3).

4.5 Quality conformance examination. Using the sampling plan and associated criteria in TABLES V and VI, the quality conformance examination shall be performed on a defect basis in accordance with the classification of defects contained in TABLE VII. For an individual defect (single characteristic), the criteria in TABLE V shall be used. For total defects

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within a major or minor class, the criteria of TABLE VI shall be used. All nonconforming material shall be removed from the lot. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification shall be used in performing visual examination for such defects as burrs and foreign matter.

TABLE VI. Total defect criteria.

Number of defect classifications in class*	Sample size			
	110	150	225	300
	Major defects allowed			
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
	Minor defects allowed			
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

*e.g. 101-119 are 19 defect classifications in a major class.

4.5.1 Reclaimed parts. Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.5.2 Reprocessed parts. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in TABLES V and VI. For an individual defect (single characteristic), the criteria in TABLE V shall be used. For total defects within a major or minor class, the criteria in TABLE VI shall be used. All nonconforming material and parts shall be rejected. All parts containing nylon pellets in which the existing protective finish is stripped and protective finish reapplied, shall have new nylon pellets inserted in accordance with the applicable drawings.

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4.6 Test methods and procedures.

4.6.1 Hydrostatic pressure. Prior to the application of the protective finish, each component listed in TABLE III shall be individually mounted in a test device, using the fixture for sealing all openings and subjected to the applicable internal hydrostatic pressure specified in TABLE III to determine compliance with 3.3. The test device used shall internally seal the components within 2.0 inches of each end without applying an axial load other than that applied by the liquid pressure. Any evidence of leakage or mechanical failure shall be cause for rejection of that component.

4.6.2 Shear pin strength. The shear pin strength test shall be performed after application of the protective finish. The pin shall be inserted in the fixture and tested in conjunction with a tensile testing machine. The radial load shall be applied at the rate of 0.125 inches per minute. Failure of any pin to shear within the range of 36 to 46 lbs. shall be cause for rejection of the lot of pins.

4.6.3 Protective finish. Test procedures and acceptance criteria for protective finish shall be in accordance with the applicable documents referenced on the drawings.

4.6.4 Residual magnetism. Each catapult assembly shall be slowly passed 6 in. in front of a MS17983 compass along the entire length of the catapult in accordance with FIGURE 1. If the compass indicator deflects more than 5° in either direction, the catapult assembly shall be rejected. The test shall be performed in an area that is shielded from local magnetic effects. When the reworking and loading of an assembly are performed by the same facility, the residual magnetism requirement shall be performed on the loaded assembly only.

4.6.5 Breakaway torque. A torque wrench set at the torque requirements specified in 3.4 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque. Any evidence of loosening or deformation of any component shall result in rejection of the catapult assembly.

4.6.6 Nozzle leak test. The retainer-nozzle ball O-ring shall be tested for leakage by partially assembling the catapult so that the retainer-nozzle subassembly and motor tube are assembled. The assembly shall then be pressurized to 40 psig and the nozzle swiveled through its operational range. Any loss of pressure within 15 seconds after nozzle has been fully swiveled shall result in rejection of the catapult assembly.

4.6.7 Release O-ring leak test. The release O-ring shall be tested for leakage by pressurizing the launcher tube-mount subassembly with nitrogen at 200 ± 5 psig. Any loss of pressure within 15 seconds of pressurization shall result in rejection of the catapult assembly.

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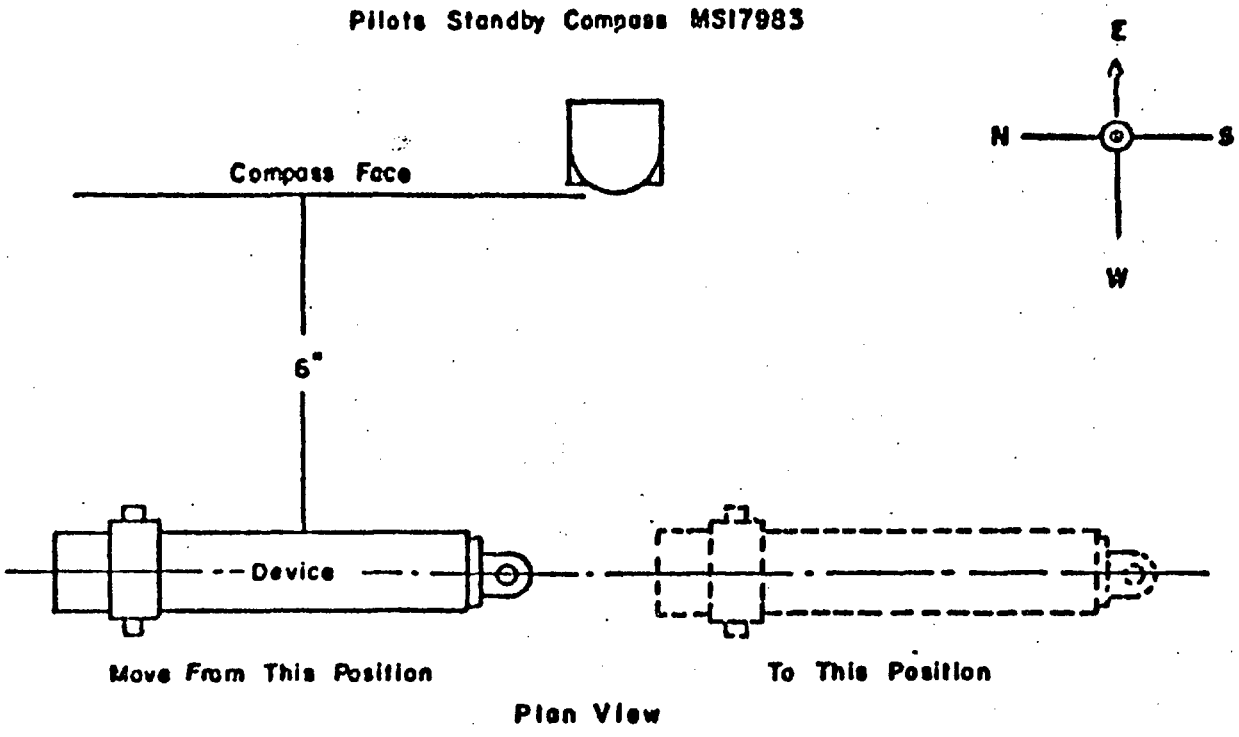


FIGURE 1. Layout for determination of residual magnetism in device.

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4.6.8 Ballistics. The test samples shall be tested to the ballistic requirements of MIL-C-48568.

4.6.9 Radiographic inspection. Rocket catapults shall be radiographically inspected in accordance with MIL-STD-453, Radiographic Quality Level II, at 0° and 90° positions. Any deficiency shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and rocket catapult components 0.03 inch in length or width, whichever is smaller.

4.7 Examinations.

4.7.1 Examination for critical defects. Each catapult assembly shall be examined for critical defects. Any catapult assembly containing one or more critical defects shall be rejected.

4.7.2 Examination for major and minor defects. Examination for major and minor defects shall be performed on a single defect basis, using the sampling plan and associated criteria in TABLE V in accordance with the classification of defects contained in TABLE VII. If the total number of defects for any classification for either major or minor defects exceeds the acceptance number specified in TABLE VI, the lot shall be rejected. All non-conforming material and parts shall be rejected. Four power (4X) magnification shall be used in performing visual examination for such defects as burrs and foreign matter.

TABLE VII. Classification of defects.

Classification	Defect	Method of inspection
Firing pin (Drawing 8596909)		
CRITICAL		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish of body and striker improper	Visual
104	Burr or sharp corner, except as required	Visual
105	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect or illegible	Visual

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TABLE VII. Classification of defects. - continued

Classification	Defect	Method of inspection
	Wire Lock, Drawing 10533781	
CRITICAL		
None defined		
MAJOR		
101	Straightness	Gage
102	Width, min.	Gage
103	Thickness, min.	Gage
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual
MINOR		
201	Width, max.	Gage
202	Thickness, max.	Gage
203	Thickness, at tang end	Gage
204	Height of tang	Gage
205	Height of chamfer end	Gage
206	Angle of tang	Gage
207	Radius or chamfer missing or incorrect	Visual
208	Surface finish improper	Visual
	Spring, Locking, Drawing 10533782	
CRITICAL		
None defined		
MAJOR		
101	Length, solid, max.	Gage
102	Load at specified heights	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual
MINOR		
201	Diameter, outside	Gage

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
Key, Drawing 10533783		
CRITICAL None defined MAJOR 101 102 103 MINOR None defined	 Protective finish inadequate or defective Foreign matter Burrs or sharp edges	 Visual Visual Visual
Lug, Drawing 10533785		
CRITICAL None defined MAJOR 101 102 103 MINOR None defined	 Protective finish inadequate or defective Burr Foreign matter	 Visual Visual Visual
Cam, Drawing 10551411		
CRITICAL None defined MAJOR 101 102 103	 Pitch diameter of thread, max. Minor diameter of thread, min. Concentricity of small radius with outside radius	 Gage Gage Gage

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Cam, Drawing 10551411	
MAJOR		
104	Concentricity of large radius with outside radius	Gage
105	Width of groove	Gage
106	Width of flange	Gage
107	Location of groove from face	Gage
108	Location of holes	Gage
109	Angles of lead of helix	Gage
110	Metal defective	Visual
111	Protective finish inadequate or defective	Visual
112	Burr	Visual
113	Foreign matter	Visual
MINOR		
201	Outside radius	Gage
202	Length from sharp edge to groove	Gage
203	Location of groove from datum	Gage
204	Angular location of holes	Gage
205	Angular width from center line	Gage
206	Angle, primary center section	Gage
207	Location to datum	Gage
208	Radius, small, center section	Gage
209	Radius, large, center section	Gage
210	Radius or chamfer missing or incorrect	Visual
211	Marking missing, incorrect or illegible	Visual
212	Surface finish improper	Visual
	Cam, Drawing 10551411 (Rework)	
CRITICAL		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect or illegible	Visual

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Head, Drawing 10551412	
CRITICAL		
1	Previous serial number stamped out	Visual
MAJOR		
101	Pitch diameter of gas intake port thread, max.	Gage
102	Pitch diameter of gas intake port thread, min.	Gage
103	Minor diameter of gas intake port thread, max.	Gage
104	Minor diameter of gas intake port thread, min.	Gage
105	Pitch diameter of internal thread, min.	Gage
106	Minor diameter of internal thread, min.	Gage
107	Metal defective	Visual
108	Protective finish inadequate or defective	Visual
109	Burr or sharp corner	Visual
110	Foreign matter	Visual
111	Diameter of mounting hole	Gage
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Mount, Drawing 10551413	
CRITICAL		
None defined		
MAJOR		
101	Pitch diameter of thread, min.	Gage
102	Minor diameter of thread, min.	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Diameter of mounting lugs	Gage

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TABLE VII. Classification of defects. - Continued

Classification	Defects	Method of inspection
MAJOR		
106	Burr or sharp corner	Visual
107	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Nozzle Drawing 10551414	
CRITICAL		
1	Thickness of wall at thin section (A-A)	Gage
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish on surfaces calling for 63 finish or better, improper	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Pin, Shear Drawing 10551415	
CRITICAL		
1	Shear test	Gage
MAJOR		
101	Diameter, min.	Gage
102	Total length, min.	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
MINOR		
201	Diameter, max.	Gage
202	Total length, max.	Gage
203	Surface finish improper	Visual
204	Radius or chamfer missing or incorrect	Visual
	Release, Drawing 10551416	
CRITICAL		
None defined		
MAJOR		
101	True position of flange with end	Gage
102	Concentricity of small diameter with end	Gage
103	Concentricity of flange end diameter on right end of view with center body diameter	Gage
104	Concentricity of groove with end diameter	Gage
105	Diameter of flange on left end of view	Gage
106	Diameter of body at center	Gage
107	Diameter of body at groove end	Gage
108	Diameter of groove	Gage
109	Thickness of wall, min.	Gage
110	Width of groove	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish of 63 or better, improper	Visual
114	Burr or sharp edge	Visual
115	Foreign matter	Visual
116	Location of flange	Gage
117	Width of O-Ring groove	Gage
MINOR		
201	Diameter of hole	Gage
202	Diameter at thin section	Gage
203	Diameter at thick section	Gage
204	Depth of hole	Gage
205	Length of chamfer	Gage
206	Total length	Gage
207	Width of flange	Gage
208	Angle of chamfer	Gage
209	Location of groove from end	Gage
210	Marking missing, incorrect or illegible	Visual
211	Radius or chamfer missing or incorrect	Visual
212	Surface finish improper, except as otherwise classified	Visual

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Release, Drawing 10551416 (Rework)	
CRITICAL		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Retainer, Drawing 10551418	
Critical		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish of 63 or better improper in spherical surface	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect, or illegible	Visual
	Retainer, Drawing 5184309	
MAJOR		
101	Angular location of cam groove ends (2 places)	Gage
102	Cam groove end radius, defective	Gage
103	Surface finish of groove, body and flanges	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish of 63 or better improper in spherical surface	Visual
MINOR		
201	Marking missing, incorrect or illegible	Visual

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Track, Drawing 10551421	
CRITICAL		
None defined		
MAJOR		
101	Concentricity of intermediate I.D. with small I.D. (C-C)	Gage
102	Inside diameter (A-A)	Gage
103	Protective finish inadequate or defective	Visual
104	Surface finish of 63 or better improper	Visual
105	Burr or sharp corner	Visual
106	Foreign matter	Visual
107	Inside diameter (C-C)	Gage
108	Inside diameter (B-B)	Gage
109	Concentricity of large I.D. with small I.D. (B-B)	Gage
110	Groove width	Gage
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Tube, Booster Assembly, Drawing 10551422	
CRITICAL		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual
105	Pellet missing, loose or improperly inserted	Manual/Visual
106	Pitch diameter of thread, max.	Gage
107	Major diameter of thread, max.	Gage
MINOR		
201	Marking missing, incorrect or illegible	Visual

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TABLE VII. Classification of defects.- Continued

Classification	Defects	Methods of inspection
	Tube, Booster, Drawing 5184310	
CRITICAL		
None defined		
MAJOR		
101	Length from end of tube to second O-ring groove (.50 REF)	Gage
102	Width of second O-ring groove	Gage
103	Depth of second O-ring groove	Gage
104	Concentricitity of second O-ring groove	Gage
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr or sharp corner	Visual
108	Foreign matter	Visual
109	Pellet missing, loose or improperly inserted	Manual/Visual
110	Pitch diameter of thread, max.	Gage
111	Major diameter of thread, max.	Gage
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Tube, Launcher, Assembly, Drawing 10551423	
CRITICAL		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish inside of tube, threaded end	Visual
104	Foreign matter	Visual
105	Major diameter of threads, max.	Gage
106	Pitch diameter of threads, max.	Gage
MINOR		
201	Marking missing, incorrect or illegible	Visual
	Tube, Motor, Drawing 10551424	
CRITICAL		
None defined		
MAJOR		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	Visual
MINOR		
201	Marking missing, incorrect, or illegible	Visual

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TABLE VII. Classification of defects. - Continued

Classification	Defect	Method of inspection
	Plate, Drawing 10551433	
CRITICAL		
None defined		
MAJOR		
101	Burr or sharp corner	Visual
102	Foreign matter	Visual
MINOR		
201	Surface finish improper	Visual
	Spring, Grain Drawing 11731754	
CRITICAL		
None defined		
MAJOR		
101	Parallelism of ends	Gage
102	Load at specified heights	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual
MINOR		
201	Diameter, inside, large end	Gage
202	Diameter, inside, small end	Gage
203	Radius missing or incorrect	Visual
204	Direction of turn of helix	Visual
205	Surface finish improper	Visual
206	Interference between coils at closed heights	Visual
	Washer, Drawing 11820367	
CRITICAL		
None defined		
MAJOR		
101	Diameter, outer	Gage
102	Protective finish	Visual
MINOR		
	All other characteristics	

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TABLE VII. Classification of defects.- Continued

Classification	Defect	Method of inspection
Catapult, Aircraft Ejection Seat,CKU-7/A, Metal Parts Assembly (Dwg 11820362)		
CRITICAL		
1	Track rotation	Manual/Visual
2	Gas Port Hole obstructed	Visual
3	Marking missing, incorrect or illegible	Visual
4	Any part missing, except as otherwise classified	Visual
5	Improper assembly	Visual
MAJOR		
None defined		
MINOR		
201	Fungus or moisture resistant varnish missing	Visual
202	Shipping plug missing	Visual
203	Protective finish inadequate or defective	Visual

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with Dwgs. 5184311 and 5184312.

6. NOTES

6.1 Intended use. The aircraft ejection seat catapult metal parts assembly composed of new, reclaimed and modified components as applicable and as described herein and when loaded with a sleeve and grain assembly MIL-S-48567, and a M270 impulse cartridge, MIL-C-14814, will produce the catapult CKU-7/A assembly. The catapult assembly is to meet the requirements of MIL-C-48568 and is intended to be used in conjunction with other propellant actuated devices in air crew emergency escape systems used in jet-propelled aircraft.

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6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Quantity required.
c. Inspection conditions when other than as specified (see 4.3).
d. The level of packaging required (see 5.1).
e. Head identification marking if other than as specified (see 4.4.1.1).

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD Form 1664) which will be documentated in the Applicable ADL (Authorized Data List).

6.3 Definition.

6.3.1 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part or evidence of leakage, the occurence of which is other than a design function.

6.4 Protective finish.

6.4.1 Remanufactured exterior components are as follows:

a. Mount, Dwg 10551413
b. Nozzle, Dwg 10551414
c. Retainer, Dwg 10551418
d. Track, Dwg 10551421
e. Motor Tube, Dwg 10551424
f. Motor Tube Head, Dwg 10551412
g. Cam, Dwg 10551411

Custodian:
NAVY - OS
AIR FORCE - 99

Preparing Activity:
NAVY - OS

Project Number:
1377-0695

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